

Final UGV Software-Validation Prototype

Detailed BOM with prices, full architecture, every power/signal connection, chassis dimensions, and feasibility decision

India procurement edition — based on selected Robu parts and current listed/quoted pricing. Prices for local consumables are practical estimates and can vary.

Final architecture: low-speed 2WD rear-wheel drive + front Ackermann steering. The laptop handles video processing, depth, SLAM/localization, planning, and Pure Pursuit. ESP32-WROOM handles low-level actuation, sensor reading, Wi-Fi commands, and safety.

Important electrical boundary: The architecture and wiring are final. Verify the delivered TT motor label is 3–6V and load-test the XL4015 at 6.00V before extended driving. TB6612FNG is appropriate for brief peak loads only; avoid prolonged stall/high-load operation.

1. Component BOM and Prices

A. Core controllers and sensors

#	Component / model	SKU	Qty	Unit ■	Total ■	Purpose
1	ESP32-WROOM-32 DevKit	835800	1	395	395	Low-level rover controller
2	ESP32-CAM, OV3660/OV2640 — confirm supplied sensor	456154	1	759	759	Wi-Fi camera stream to laptop
3	ESP32-CAM-MB Micro-USB download module	1150750	1	109	109	Flashes ESP32-CAM firmware
4	MPU-6050 IMU breakout	2846	1	151	151	I2C acceleration/gyro sensing
5	HC-SR04 ultrasonic sensor	2444	4	69	276	Front/left/right/rear independent safety sensing
6	MG90S 180° steering servo	1308083	1	125	125	Turns front steering linkage
7	1kΩ 1/4W resistor	local/Robu	10	2	20	Four 5V-to-3.3V Echo dividers + spares
8	2kΩ 1/4W resistor	local/Robu	10	2	20	Four Echo dividers + spares

B. Drive, power, and safety

#	Component / model	SKU	Qty	Unit ■	Total ■	Purpose
9	Pair: Microbit TT DC motors, 1:48, 3–6V, 200 RPM at 6V	1121374	1 pair	446	446	Left and right rear-wheel propulsion
10	TB6612FNG 2×1.2A driver breakout (DFRobot Fermion option)	Robu product	1	489	489	PWM/direction drive for two motors
11	XL4015 5A buck with heatsink	1510171 / R224623	1	71	71	Adjust to 6.00V; motor rail
12	LM2596S adjustable buck, 3A	11548	1	44	44	Adjust to 5.00V; electronics rail
13	Samsung INR18650-35E 2S1P, 7.4V, 3500mAh, 2C battery	1366608	1	1075	1075	Main battery; 8.4V max; 7A stated continuous
14	5A automotive blade fuse + inline holder	local/automotive	1 set	80	80	Main battery positive protection
15	NC emergency-stop switch, DC-rated ≥5A	local/Robu	1	120	120	Independent physical power cut
16	KCD1-101 rocker switch — only if DC current rating is adequate	R132455	1	6	6	Normal master ON/OFF
17	Power distribution terminal block, positive + ground	local	1	100	100	Cleanly divides input power / common ground
18	5V 2–3A polyfuse or inline fuse	local	1	40	40	Recommended electronics-rail protection

C. Mechanics, filtering, and wiring

#	Component	Qty	Estimated ■	Purpose
19	65–70mm TT/BO compatible wheels	4	150	Two rear drive wheels + two front steering wheels

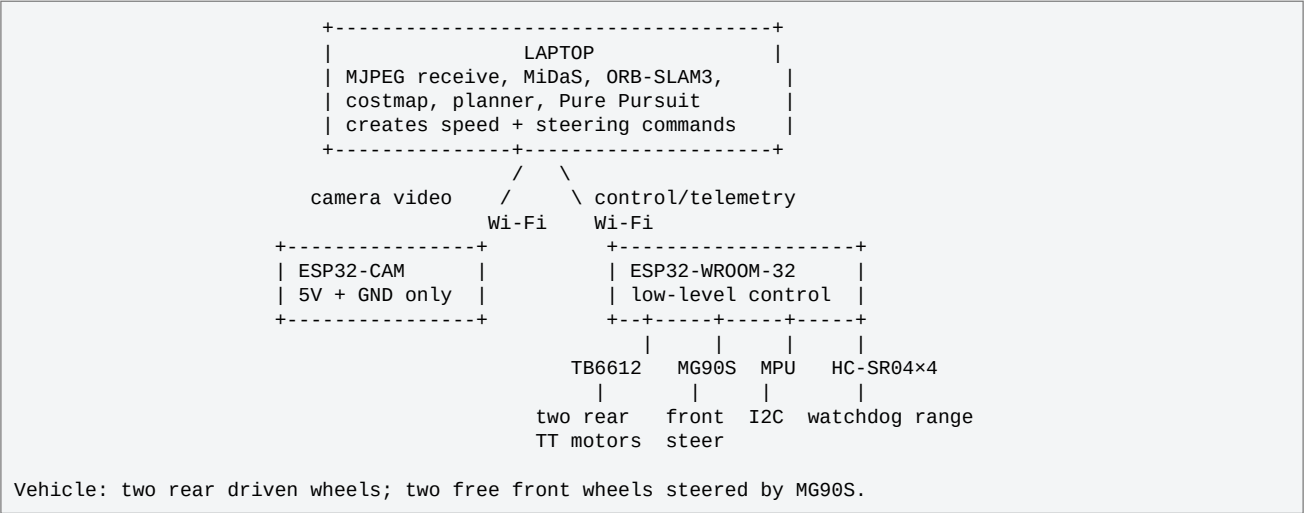
#	Component	Qty	Estimated ■	Purpose
20	3–5mm acrylic sheets	2	400	Top and bottom chassis plates
21	M3 screws, nuts, washers and standoffs	1 set	200	Assembly and mounting
22	Servo horn, tie rods, steering knuckles, front axles/bearings	1 set	200	Ackermann steering mechanism
23	470µF 16V electrolytic capacitors	2	30	5V and 6V rail filtering
24	100µF 10–16V electrolytic capacitors	2	20	5V rail filtering
25	0.1µF 50V ceramic capacitors	8	30	Two at motors; spares/decoupling
26	Perfboard about 5×7cm	1	50	Echo dividers + capacitors
27	18–20 AWG stranded wire (~3m)	1 lot	120	Battery/buck/motor high-current paths
28	22–24 AWG stranded wire (~3m)	1 lot	80	GPIO, I2C, sensor wiring
29	Dupont M-M/M-F/F-F set	1 set	150	Signal prototyping only
30	Heat shrink + zip ties	1 set	120	Insulation and strain relief
31	25mm-square checkerboard print	1	50	Camera calibration

Estimated total: ■5,816 excluding shipping and any higher-quality steering knuckles/bearings. The total uses the explicitly selected Samsung 2S battery and DFRobot TB6612FNG price; a generic TB6612FNG may be cheaper, but buy a genuine module with a proper pin breakout.

The TT motor page specifies 3–6V, 200±10% RPM at 6V, 1:48 ratio, 150mA no-load current, and 0.8kg·cm stall torque. It does not give stall current in the listing excerpt, so do not assume a numerical stall-current value without checking the delivered item or seller confirmation. The selected battery listing specifies 7.4V, 3500mAh, 2C, 8.4V max, DC charge port, and Amass Nylon-T discharge port. The XL4015 listing specifies 4–38V input, adjustable output, and 0–5A claimed output.

[web:197][web:199][web:194]

2. Complete Architecture



Wi-Fi interpretation: ESP32-CAM → laptop video and laptop ↔ ESP32-WROOM command/telemetry are two independent logical data paths. They may share the same access point and RF channel. If the control path times out, ESP32 must stop the motors even if camera video is still streaming.

3. Full Power Wiring — Every Connection

3.1 Battery positive chain

Battery + (red, 7.4V nominal / 8.4V full)

- > 18-20 AWG red -> inline fuse holder -> 5A automotive blade fuse
- > 18-20 AWG red -> E-stop NC terminal 1
- > E-stop NC terminal 2 -> rocker-switch input terminal
- > rocker-switch output terminal -> POSITIVE POWER BUS
- > Positive bus branch 1 -> XL4015 VIN+
- > Positive bus branch 2 -> LM2596 VIN+

Place the fuse physically close to the battery positive. The E-stop comes before both buck converters. Pressing it removes all power from the rover: motors, both ESP32 boards, servo, and sensors.

3.2 Battery ground / common-ground bus

Battery - (black) -> COMMON GROUND BUS

- > XL4015 VIN- and VOUT-
- > LM2596 VIN- and VOUT-
- > TB6612FNG GND
- > ESP32-WROOM GND
- > ESP32-CAM GND
- > MG90S brown/black wire
- > HC-SR04 GND x4
- > MPU-6050 GND
- > negative pins of all electrolytic capacitors

Common ground is mandatory. It is not optional. The ESP32 PWM/direction signals will be unreliable if TB6612 and ESP32 grounds are not joined.

3.3 6V motor-power branch

Connection	Exact wiring
XL4015 input	Positive bus → XL4015 VIN+; ground bus → XL4015 VIN-
XL4015 adjustment	Before connecting TB6612, use multimeter and adjust XL4015 output to exactly 6.00V
XL4015 output	XL4015 VOUT+ → TB6612 VM using 18-20 AWG red; XL4015 VOUT- → common ground using 18-20 AWG black
Motor bulk capacitor	470µF capacitor: positive leg → TB6612 VM / XL4015 VOUT+; negative leg → TB6612 GND / common ground. Keep leads short.
Motor suppression	Solder one 0.1µF ceramic capacitor directly across the two terminals of each TT motor.

3.4 5V electronics-power branch

Connection	Exact wiring
LM2596 input	Positive bus → LM2596 VIN+; ground bus → LM2596 VIN-

Connection	Exact wiring
LM2596 adjustment	Before connecting electronics, use multimeter and set output to exactly 5.00V
5V distribution	LM2596 VOUT+ → 5V rail/distribution point; LM2596 VOUT– → common ground
5V loads	5V rail → ESP32-WROOM 5V/VIN; ESP32-CAM 5V; MG90S red; HC-SR04 VCC x4
5V capacitors	470μF and 100μF capacitors: positive legs → 5V rail; negative legs → common ground; place near servo/ESP32 distribution

4. Full Signal Wiring — Each Pin

4.1 ESP32-WROOM → TB6612FNG

TB6612FNG pin	Connect to	What it does
VM	XL4015 VOUT+ (6.00V)	Motor supply; do not connect VM to ESP32 3.3V
VCC	ESP32 3.3V	Logic supply only
GND	Common ground bus	Power + GPIO reference
STBY	ESP32 GPIO23	LOW: disable H-bridge; HIGH: enable. Firmware safety control
PWMA	ESP32 GPIO13	Left motor LEDC PWM
AIN1	ESP32 GPIO26	Left direction input 1
AIN2	ESP32 GPIO27	Left direction input 2
AO1	Left TT motor wire 1	Left motor terminal
AO2	Left TT motor wire 2	Left motor terminal
PWMB	ESP32 GPIO14	Right motor LEDC PWM
BIN1	ESP32 GPIO25	Right direction input 1
BIN2	ESP32 GPIO4	Right direction input 2
BO1	Right TT motor wire 1	Right motor terminal
BO2	Right TT motor wire 2	Right motor terminal

If a wheel turns backwards when the firmware commands forward, swap only that motor two wires at AO1/AO2 or BO1/BO2. STBY is a direct H-bridge disable controlled by firmware; it is not a substitute for the physical E-stop.

4.2 ESP32-WROOM → MG90S Servo

Servo wire	Connect to	Meaning
Brown/black	Common ground bus	Ground
Red	5V rail	Servo supply; not ESP32 3.3V
Orange/yellow	ESP32 GPIO18	Steering PWM signal

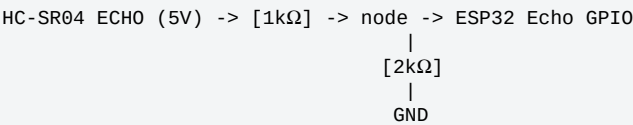
4.3 ESP32-WROOM → MPU-6050

MPU-6050 pin	Connect to	Meaning
VCC	ESP32 3.3V	3.3V-compatible IMU supply
GND	Common ground bus	Ground
SDA	ESP32 GPIO21	I2C data
SCL	ESP32 GPIO22	I2C clock
AD0	GND	Default address 0x68
INT	Leave unconnected initially	Optional future interrupt

4.4 ESP32-WROOM → Four HC-SR04 Sensors

Sensor	VCC	GND	TRIG	ECHO
Front	5V rail	Ground bus	GPIO5	via divider → GPIO16
Left	5V rail	Ground bus	GPIO17	via divider → GPIO32
Right	5V rail	Ground bus	GPIO19	via divider → GPIO34
Rear	5V rail	Ground bus	GPIO33	via divider → GPIO35

Build FOUR identical Echo voltage dividers on perfboard:



Front: GPIO16 Left: GPIO32 Right: GPIO34 Rear: GPIO35

GPIO34 and GPIO35 are input-only pins, which is correct for Echo signals. Trigger only one HC-SR04 at a time; wait for the prior echo to finish before triggering the next sensor to reduce cross-talk.

4.5 ESP32-CAM

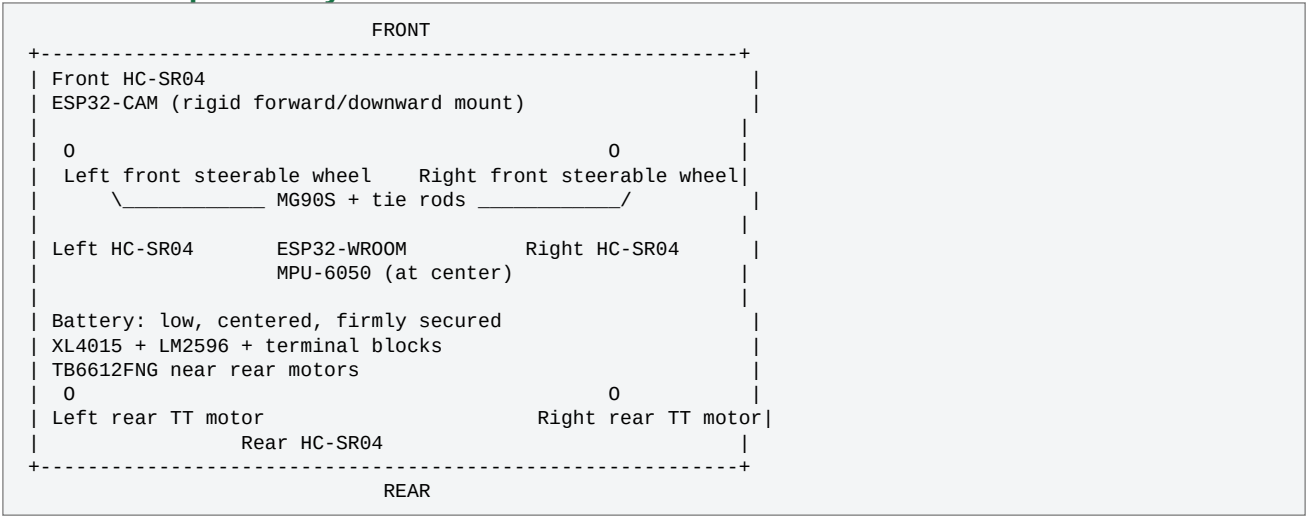
ESP32-CAM pin	Connect to	Meaning
5V	5V rail	Regulated camera-board supply
GND	Common ground bus	Ground

There is no WROOM-to-CAM GPIO connection. Program ESP32-CAM using ESP32-CAM-MB, then power the camera board from the 5V rail and connect it to Wi-Fi.

5. UGV Mechanical Dimensions

Parameter	Final starting value	Reason
Overall chassis length	300 mm	Space for front steering, battery, controllers, sensors
Overall chassis width	200 mm	Stable but compact acrylic platform
Wheelbase, front axle to rear axle	220 mm	Use in bicycle/Pure Pursuit model
Track width, wheel-center to wheel-center	180 mm	Stability and steering geometry
Wheel diameter	65–70 mm	TT/BO compatible
Ground clearance	30–40 mm	Avoids mild terrain scraping
Acrylic thickness	3–5 mm	Lightweight structural plates
Deck separation	40–60 mm	Battery/wiring/motor clearance
Max steering angle	~25–30°	Servo + linkage practical range
Minimum turning radius	~400–470 mm	$R_{min} = L / \tan(\delta_{max})$, $L=220\text{mm}$

Recommended top-view layout



6. Feasibility Decision

Area	Verdict	Why
Architecture	Feasible	Laptop runs the heavy AI/SLAM/planning; ESP32 runs real-time low-level safety/actuation
Kinematics	Feasible	Rear drive + front steering directly matches bicycle model and Pure Pursuit
Motors	Feasible for prototype	Two small TT motors are adequate for a light (~0.75 kg target) rover on flat/mild terrain
Motor driver	Conditional	TB6612 works at light normal load but should never be held in prolonged stall near 1.5A/channel
Battery	Feasible	3500mAh × 2C = 7A stated continuous; check connector and delivered label
6V buck	Conditional	XL4015 claimed 5A is appropriate, but verify 6.00V stability/temperature under actual motor load
5V buck	Feasible	A genuine 3A LM2596-class rail has margin for ESP32s, servo, and sensors
Safety	Feasible	E-stop + fuse physical layer; STBY + watchdog + ultrasonic firmware layer
Dimensions	Feasible	300×200mm and 220mm wheelbase offer room, stability, and reasonable turning radius

Final conclusion: Yes, these components and this architecture are feasible for your stated goal: validating the autonomy software on a slow, lightweight UGV. This is not intended for deep grass, sand, steep slopes, heavy payloads, or prolonged motor stalls. After bench verification of the 6V rail, fuse, and motor label, stop redesigning and build the prototype.

7. Mandatory Safe Boot/Stop Logic

```
BOOT: STBY LOW + PWM=0
-> initialize known GPIO direction states
```

```
-> initialize MPU-6050 and ultrasonic sensors
-> establish Wi-Fi
-> receive a fresh valid command
-> verify safety readings healthy
-> STBY HIGH, then permit PWM

ON COMMAND TIMEOUT / INVALID COMMAND / UNSAFE SENSOR / FAULT:
-> PWM = 0
-> STBY LOW
-> do not resume until a fresh valid command arrives

ON PHYSICAL E-STOP:
-> entire rover loses power
-> reboot and reconnect required; never resume old motion automatically
```

Test in this order: confirm 5.00V and 6.00V with a multimeter; check common-ground continuity; test IMU; test one ultrasonic divider then all four sequentially; test servo with wheels lifted; test STBY; test each motor at low PWM; test Wi-Fi timeout and obstacle stops; then mount the complete chassis.